

Innovation Activities and Integration through Vertical Acquisitions

Laurent Frésard

Universita della Svizzera italiana (USI), Swiss Finance Institute

Gerard Hoberg

Marshall School of Business, University of Southern California

Gordon M. Phillips

Tuck School at Dartmouth College and National Bureau of Economic Research

We examine the determinants of vertical acquisitions using product text linked to product vocabulary from input-output tables. We find that the innovation stage is important in understanding vertical integration. R&D-intensive firms are less likely to become targets of vertical acquisitions. In contrast, firms with patented innovation are more likely to sell to vertically related buyers. Firms' R&D intensity is a more important deterrent to their vertical acquisitions when the provision of innovation incentives by potential acquirers is more difficult. The role of patents in fostering vertical acquisitions is more prevalent when potential buyers face a higher risk of holdup. (*JEL* G32, G34, L22, L25, O34)

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The scope of firm boundaries and whether to organize transactions within the firm (integration) or by using external contracting is of major interest in

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understanding why firms exist. A large literature investigates the determinants of vertical integration and, more recently, the relationship between firms' vertical organization and innovation activities.¹ Yet, existing research remains largely silent on the role of firms' innovation activities in explaining vertical acquisitions. In this paper, we provide novel evidence that the stage of development of innovative assets is a strong determinant of firms' vertical acquisitions and the boundaries of the firm.

Our analysis builds on the theories of the firm of Williamson (1971), Grossman and Hart (1986), and Hart and Moore (1990) who emphasize that, because of incomplete contracting and the inalienability of human capital, the vertical boundaries between firms (separation or integration) are determined by trading off their ex ante investment incentives and ex post opportunistic behaviors. This trade-off is regulated by the allocation of control rights over the assets that are specific to firms' vertical relationships, as the firms that formally control these assets have more incentives to invest ex ante due to increased bargaining power ex post. Therefore, change in control – vertical acquisitions – should occur when the relative importance of each party's incentives to add value to the overall relationship shifts.²

Early stage innovation activities (i.e., unrealized innovation) are particularly sensitive to the allocation of control rights because investments in innovation are typically not fully contractible, often unverifiable, heavily depend on human capital, and are subject to holdup because of their specificities (e.g., Acemoglu 1996). When innovative assets require further investment and development, it is optimal to leave control to the firms that perform the innovation, as their incentives are most important for the value of the vertical relationship (e.g., Aghion and Tirole 1994), and because their employees may leave in case of acquisition and take the unrealized innovation (i.e., their ideas) with them (e.g., Hart and Moore 1994).³ Vertical separation preserves incentives for the innovator (Grossman and Hart 1986). In contrast, when innovation is realized, and protected by legally enforceable patents, incentives to invest in innovation decline in importance and incentives to commercialize the innovation increase. In this context, vertical acquisitions optimally reallocate control to the firms that commercialize the innovation and thus integration reduces the risk of holdup by the other party (Williamson 1971, 1979; Klein, Crawford, and Alchian 1978). We thus predict that acquisitions by vertically related buyers are less likely when firms' innovation is still unrealized, but more likely when it is realized and patented.

¹ See Lafontaine and Slade (2007) and Bresnahan and Levin (2012) for recent surveys.

² Holmstrom and Milgrom (1991) and Holmstrom and Milgrom (1994) also emphasize the role of incentives in firm structure. Gibbons (2005) summarizes the large literature and highlights that the costs and benefits of vertical integration depend on transactions costs, rent seeking, contractual incompleteness, and the specificity of the assets involved in transactions.

³ For instance, the original innovating team could freely form a new firm and use the unrealized innovation to compete against the purchasing firm.

To test this hypothesis, we construct a novel measure of vertical relatedness between firm-pairs by linking product vocabularies from the Bureau of Economic Analysis (BEA) input-output tables to firms' 10-K product descriptions filed with the Securities and Exchange Commission (SEC). Because 10-Ks are updated annually, the result is a dynamic directed network of vertical relatedness between all publicly traded firms between 1996 and 2013. In this text-based network, we identify for instance that Tesla is vertically related (downstream) to Honeywell, Goodyear, and Sun Hydraulics in 2010. This pairwise network allows us to identify precisely which mergers and acquisitions occur between firms offering vertically related products. We use the same approach to also measure the degree of firms' vertical integration based on whether firms use product vocabulary that spans vertically related markets. Using a large sample of more than 8,000 firms, we find a strong association between the stage of development of firms' innovation and vertical acquisitions.

Our main tests focus on targets in vertical deals since they are the party that relinquishes control rights by selling, and for which the trade-off between ex ante investment incentives and holdup is important. Consistent with vertical separation being optimal when innovation is unrealized, we find that R&D-intensive firms are significantly less likely to be acquired in vertical transactions. In contrast, when innovation is realized and protected by legally enforceable patents, firms are more likely to be acquired by a vertical buyer. The positive link between firms' patenting intensity and their vertical acquisitions is consistent with patenting marking a shift away from incentives for further R&D and toward incentives to commercialize the innovation.

The opposite roles of unrealized and realized innovation in vertical acquisitions are highly robust across a host of alternative specifications, including alternative measures of firms' innovative efficiency and strict controls for the potential multicollinearity between firms' R&D and patenting intensity. The estimated relations are also economically important as a 1-standard-deviation increase in firms' R&D intensity is associated with a large 0.46% percentage point reduction in their probability of being acquired in a vertical transaction. This decrease represents roughly one-third of the unconditional probability of being acquired by a vertical buyer. The economic magnitude of firms' patenting intensity in explaining vertical acquisitions are even larger.

Western Digital's purchase of Komag in 2007 illustrates the logic of our results. Komag specialized in developing thin-file disks of the sort used in hard drives. Buying Komag transferred control of Komag's realized innovation to Western Digital, thus reducing ex post holdup and increasing Western Digital's commercialization incentives. Western Digital's CEO John Coyne commented that "the acquisition of Komag will enable Western Digital to optimize synergies through the integration of heads and media" (Duncan 2007). Another example is GE's acquisition of Edwards Systems Technology in 2004, which gave GE control of Edward's fire detection technology, used in GE's broad security offering. Will Woodburn, CEO of GE Infrastructure, indicated that Edwards'

technology will enable GE to offer an “integrated facility management solution to partners and end users” (Buildings 2004).

Our hypothesis specifically points to the trade-off between maintaining innovation incentives and limiting holdup risk. We thus examine whether the negative relation between R&D and acquisition, and the positive association between patenting and acquisition, is stronger when these trade-offs are most salient. First, we show that the negative link between firms’ R&D intensity and vertical acquisitions is significantly stronger when potential vertical buyers likely find it difficult to maintain the innovation incentives of the target and keep its key employees post-acquisition, as measured using inventor mobility and the intensity of each firm’s 10-K disclosures relating to employee incentives.⁴ Second, we show that the positive link between firms’ patenting intensity and vertical acquisitions is significantly stronger when opportunistic behaviors are more likely, which we measure using patent infringement litigation, poor secondary market liquidity, and high product market concentration.

Our tests are consistent with the idea that investment incentives and holdup risk increase the sensitivity of vertical boundaries to the stage of innovation. Nevertheless, our tests do not fully rule out some alternative explanations and we further test two specific alternatives. First, our results could be driven by potential vertical buyers relying on patent grants as signals of a target’s innovation efficiency. Second, our findings might reflect firms responding to the likelihood of vertical acquisitions by reducing their R&D intensity and increasing their patenting intensity (reverse causality) to attract a better offer. Several tests mitigate both concerns. For instance, inconsistent with patent grants signaling innovative efficiency, our results are robust to including controls for firms’ innovative efficiency (patents by dollar of R&D). We further examine the timing of vertical acquisitions and find that patenting intensity increases around the acquisition. The drop in firms’ R&D intensity occurs mostly after control is transferred to the buyers, which is inconsistent with a reverse causality explanation.

Inconsistent with many other alternatives, we show that our results are unique to vertical acquisitions as we find opposite or insignificant results for nonvertical acquisitions (in line with Bena and Li 2014). We recognize that part of our results might arise because some acquirers might use patent grants as an indication of the degree of the target’s success, triggering their acquisition. This channel can be distinct from our explanation if no *ex ante* incentives are required for the effort needed to achieve that degree of success. Yet, our

⁴ Note we do not study “acqui-hires,” where firms acquire a company to recruit its employees without necessarily showing an interest in its current products and services. In many such cases, the product that these previously separate employees were working on either has failed or is not significant enough to develop within the firm.

subsample tests based on proxies for incentive provision and holdup risk provide specific support for our proposed incentive-based channel.

Finally, we show that our new measures of the degree of firms' vertical integration are also consistently related to the stage of innovation development. In particular, firms operating in industries characterized by high levels of R&D intensity display lower levels of vertical integration. In contrast, firms in patenting intensive industries appear significantly more integrated vertically, in line with vertical integration being more prevalent in sectors where innovation is realized. Using within firm analysis, we uncover a positive relationship between vertical integration and firms' patenting intensity over time, but only for firms that have been vertical acquirers. In contrast, changes in firms' degree of vertical integration is largely unrelated to their R&D intensity. These results support our central hypothesis as these firms are specialized (i.e., nonintegrated) and we predict and find that they tend to exit the sample via vertical acquisitions once their innovation becomes realized.

Our findings contribute to the large literature examining the determinants of firms' vertical boundaries, and in particular to recent papers linking vertical integration to innovation and intangible assets. Acemoglu et al. (2010) show, in a sample of UK manufacturing firms, that backward integration is positively (negatively) related to the R&D intensity of the downstream (upstream) industry. Using Census data, Atalay, Hortacsu, and Syverson (2014) report limited physical shipments within vertically integrated firms in the United States, suggesting that innovation and intangible capital (which do not require shipments) might be responsible for firms' vertical organization.⁵ By showing that firms' vertical boundaries are related to the stage of innovative development, our paper provides direct evidence of the importance of innovation for firms' vertical boundaries.

Our paper also adds to the literature on the determinants and implications of vertical acquisitions. Fan and Goyal (2006) and Kedia, Ravid, and Pons (2011) examine stock market reactions to vertical deals, and Ahern (2012) shows that the division of stock market gains is determined in part by customer or supplier bargaining power. Ahern and Harford (2014) examine how supply chain shocks translate into vertical merger waves. Our study is distinct and focuses on the stage of innovation and a novel hypothesis linking this to the choice of firms' organizational form. Our results are consistent with the view that vertical acquisitions emerge as an optimal way to transfer the residual rights of control of relationship-specific intangible assets to the party whose investment incentives are most important for the success of the relationship, thus also reducing holdup risk. This motive is distinct from existing acquisition

⁵ The authors document a decline in nonproduction workers in vertically acquired establishments. They also find a shift in production of the acquirer's products toward the acquired firms' establishments.

motives emphasized in existing research including neoclassical theories, agency theories, and horizontal theories.⁶

Our focus is on the motives for acquisitions of vertically related firms who have actual products (i.e., publicly listed firms), and we do not directly consider licensing or patent purchases, as they are outside the scope of our study.⁷ We conjecture that licensing occurs when follow-on integration or tailoring of the technology to a particular use after contracting is less important. Patent sales occur when the scope of the patent is well defined and the acquirer does not need resources or employees from the target to help integrate the product or further develop the technology. Our hypothesis differs from just purchasing patents or licensing because acquisitions among more complex publicly traded firms typically involves additional needs beyond the associated patents alone. For example, many innovation-driven acquisitions require continued participation by the targets' employees either regarding future technological development or in adapting the technology to the purchaser.⁸ Thus, we expect to observe technology-driven vertical acquisitions despite the possibility of patent purchases or licensing.

A key methodological contribution is that we can identify vertical relatedness at the detailed firm-pair level. Existing measures are based on static NAICS industry classifications and not only fail to provide firm-specific measures, but are slow to update over time and are based on production processes and not the products and services offered by firms.⁹ Our textual measures do not rely on the quality of the Compustat segment tapes to delineate firms' product markets, the quality of the NAICS classification, or the links between NAICS codes and the Input-Output tables, which are based on BEA classifications. This methodology for measuring vertical linkages between firms extends the work of Hoberg and Phillips (2016), who focus on horizontal links using 10-K text. Our approach thus contributes to the growing literature that uses text as data in finance and economics, recently surveyed by Gentzkow, Kelly, and Taddy (forthcoming).

1. Vertical Acquisitions and the Stage of Innovation

In this section, we discuss a framework to motivate our tests based on the analysis of Grossman and Hart (1986), Aghion and Tirole (1994), and Williamson (1971). As an illustration of the contrasting effects of

⁶ See Maksimovic and Phillips (2001), Jovanovic and Rousseau (2002), and Harford (2005) for neoclassical and *q* theories; see Morck, Shleifer, and Vishny (1990) for an agency motivation for acquisitions; and see Phillips and Zhdanov (2013) for a recent horizontal theory of acquisitions.

⁷ For studies in these areas, see Teece (1986) and Arora and Fosfuri (2003) for licensing and Serrano (2010) for patents sales and transfers.

⁸ Apple, for example, purchased Intrinsity for its technology and the engineers who developed low power fast chips at Intrinsity further enhanced Apple's mobile and iPad chips.

⁹ The Federal Register states "NAICS was developed to classify units according to their production function. NAICS results in industries that group units undertaking similar activities using similar resources but does not necessarily group all similar products or outputs" (Federal Register 2014).

realized and unrealized innovation on firms' vertical boundaries, consider two vertically related firms (e.g., upstream supplier and a downstream producer) that cooperate to offer a product or service. These firms can operate as separate entities or combine into a vertically integrated firm.¹⁰

The joint value of their vertical relationship and the price they charge to consumers depends positively on R&D product investments and commercialization investments, both of which are specific to the relationship. Both upstream and downstream firms can conduct R&D. While both firms conduct R&D, it is relatively more likely (but not a given as our examples later show) that upstream firms conduct product R&D and downstream firms that are closer to the end-consumer conduct investments and R&D that is useful for commercialization. Either firm can have a differential mix of realized and unrealized innovation.

As is standard in the literature (e.g., Aghion and Tirole 1994), the outcome of the product innovation effort in each period is risky, succeeding only probabilistically, and is noncontractible and unverifiable. The outcome of innovation effort is protected by legally enforceable patents in case of success, but remains intangible if unsuccessful. The commercialization investment in a given period is less risky than the innovation effort, and occurs after its outcome is realized. As the product (or service) resulting from firms' relationship matures over time, the marginal value of innovation effort to add new technological features naturally decreases, whereas that of commercialization effort marginally increases.

In our context, the optimal organizational form depends on the stage of development of the innovation. First, consider the firms operating separately. It is optimal for firms to stay separate when one firm's product innovation efforts have not succeeded yet and further innovation effort thus benefits the relationship. In that case, separation optimally allocates control rights to the party whose investment incentives are relatively more important. Indeed, transferring control rights to the other firm when innovation is still unrealized lowers innovation incentives, as innovative effort is not fully contractible. In addition, integration does not guarantee the acquirer's access to the other firm's unrealized innovation because of the absence of clearly defined property rights. For instance, the value of unrealized innovation could be encapsulated in employees' knowledge, which is inalienable, as in Hart and Moore (1994), and employees might freely leave in the case of integration. Intuitively, staying separate is optimal when further incentives for R&D benefit the overall relationship. In that case, individual ownership maintains *ex ante* incentives for the firms to further invest. Separation optimally allocates residual rights of control to the party who has unrealized innovation where incentives are more important.

¹⁰ We included a simple model developing these ideas in an earlier version of this paper. This model is available on the authors' Web sites.

Now consider the integration decision. At each point in time, the firms can either operate as separate entities or decide to integrate through one party (either upstream or downstream) buying the other.¹¹ The party that sells its assets (the target) loses control rights over the assets sold, and makes less new product R&D investment, as this investment is noncontractible and the parties would not be able to ensure payment ex post. The realization of R&D and the grants of patents are thus key to determining whether firms integrate or remain separate.

When the innovation is successful and its features are protected by patents, incentives for further innovation decline because of the decreasing marginal value of innovation effort for the overall relationship. In contrast, the incentives of the potential acquirer to invest in commercialization marginally increase. However, the crucial tension is that without legal control rights on the outcome of the realized innovation through the ownership of the patents, the user of the innovation faces a risk of holdup, in the spirit of Williamson (1971), whereby one firm can threaten to holdup the other firm after significant investment has been made by both parties. The acquisition decision is made to encourage ex post commercialization investments and to protect the two parties' investments in the relationship after more of the innovation becomes realized through patents.

This holdup risk is likely in production when commercialization is more important. Commercialization investments can include investments that lower the cost of the product. Holdup risk can also include supply chain problems that can persist when firms are separate and the quality of the engineers or people involved in the production of the components supplied cannot be easily contracted upon.¹² To encourage commercialization incentives, the overall relationship should optimally allocate the residual rights of control to one party to reduce holdup costs. At this stage, vertical integration through acquisition is optimal as it maximizes total surplus.

The trade-off between ex ante innovation incentives and ex post holdup risk leads to our central prediction:

Central Prediction: *Firms are less likely to be acquired by a vertically related buyer when their innovation is unrealized. Firms are more likely to be acquired by a vertically related buyer when their innovation is realized and protected by patents.*

¹¹ Note that the overall framework is agnostic about whether the acquirer is the upstream firm or the downstream firm closer to the customer. Several examples show that the acquirer can be either the upstream firm or the downstream firm. Western Digital, a disk drive manufacturer, is an example of a downstream firm buying an upstream firm. They purchased Komag Inc., a producer of thin-film discs used in hard drives. We also note that Conexant, a semiconductor manufacturer, acquired an upstream firm, Globespan Virata Inc., a maker of broadband communications products that use semiconductor chips. In both of these cases, the target was a high R&D firm that, over time, realized more patents.

¹² An example would be Boeing purchasing some of its previously separate suppliers and integrating them given supply chain problems. For additional examples, see von Simson (2010).

We test this prediction using new text-based measures of vertical relatedness between firms, and by examining the impact of firms' R&D and patenting intensity on their likelihood of being purchased in vertical transactions.

2. Measuring Vertical Relatedness

This section develops and validates our text-based measure of vertical relatedness.

2.1 Data from 10-K business descriptions

We start with all firm-year observations in Compustat from 1996 to 2013 with sales of at least \$5 million and positive assets.¹³ We follow the same procedures as Hoberg and Phillips (2016) to identify, extract, and parse 10-K annual firm business descriptions from the SEC Edgar database. These 10-Ks are merged with the Compustat-based sample using the central index key (CIK) mapping to gykey provided in the WRDS SEC Analytics package. Item 101 of Regulation S-K requires business descriptions to accurately report (and update each year) the significant products firms offer. We obtain 92,087 firm-years in the merged Compustat/Edgar universe.

2.2 Data from input-output tables

We then use the 2002 BEA input-output (IO) tables, which provide dollar flows between producers and purchasers in the U.S. economy (including households, government, and foreign buyers of U.S. exports). The IO tables are based on two primitives: "commodity" outputs (any good or service) defined by the Commodity IO Code, and producing "industries" defined by the Industry IO Code. In 2002, there were 424 commodities and 426 industries in the "Make table," which reports the dollar value of each commodity produced by each industry. There are 431 commodities purchased by 439 industries or end users in the "Use table," which reports the dollar value of each commodity purchased by each industry.¹⁴ We use these IO tables to compute three items: (1) commodity-to-commodity (upstream to downstream) correspondence matrix (V), (2) commodity-to-word correspondence matrix (CW), and (3) commodity-to-"exit" correspondence matrix (E).

In addition to the numerical values in the BEA data, we use an often overlooked resource: the "Detailed Item Output" table, which verbally

¹³ We show in the Online Appendix that our results are robust to any minimum sales thresholds ranging from \$1 million to \$10 million.

¹⁴ An industry can produce more than one commodity: in 2002, the average (median) number of commodities produced per industry was 18 (13). Industry output is also concentrated as the average commodity concentration ratio is 0.78. Costs are reported in both purchaser and producer prices. We use producer prices. Seven commodities in the Use table are not in the Make table, for example, compensation to employees. Thirteen "industries" in the Use table are not in the Make table. These correspond to "end users" and include personal consumption, exports and imports, and government expenditures.

Table 1
BEA vocabulary example: Photographic and photocopying equipment

Description of commodity subcategory	Value of production (\$Mil.)
Still cameras (hand-type cameras, process cameras for photoengraving and photolithography, and other still cameras)	266.1
Projectors	72.4
Still picture commercial-type processing equipment for film	40.5
All other still picture equipment, parts, attachments, and accessories	266.5
Photocopying equipment, including diffusion transfer, dye transfer, electrostatic, light and heat sensitive types, etc.	592.4
Microfilming, blueprinting, and white-printing equipment	20.7
Motion picture equipment (all sizes 8 mm and greater)	149.0
Projection screens (for motion picture and/or still projection)	204.9
Motion picture processing equipment	23.0
Processed commodity vocabulary: microfilming, whiteprinting, blueprinting, interchangeable, film, projectors, rear, viewers, screen, mm, photoengraving, photolithography, cameras, photographic, motion, electrostatic, diffusion, dye, heat, projection, screens, picture, still, photocopying	

This table provides an example of the BEA commodity “photographic and photocopying equipment” (IO commodity code #333315). The table displays its subcommodities and their associated product text, along with the value of production for each subcommodity.

describes each commodity and its subcommodities. The BEA also provides the dollar value of each subcommodity’s total production and a commodity’s total production is the sum of these subcommodity figures.¹⁵ Each subcommodity textual description uses between 1 to 25 distinct words (the average is 8) that summarizes the nature of the good or service provided. Table 1 contains an example of product text corresponding to the BEA “photographic and photocopying equipment” commodity (IO commodity code #333315). We label the complete set of words associated with a commodity as “commodity words.”

We follow Hoberg and Phillips (2016) and only consider nouns and proper nouns. We then apply four additional screens to ensure our identification of vertical links is conservative. First, because commodity vocabularies identify stand-alone product markets, we manually discard expressions that indicate a vertical relation, such as “used in,” “made for,” or “sold to.” Second, we remove expressions indicating exceptions (e.g, phrases beginning with “except” or “excluding”). Third, we discard common words from commodity vocabularies.¹⁶ Fourth and finally, we remove any words that do not frequently coappear with the other words in a given commodity’s vocabulary. This ensures that horizontal links are not mislabeled as vertical links. To do so, we compute the fraction of times each focal word coappears with the same peer words used in a given IO commodity when it appears in a 10-K business description using all 10-Ks in 1997. For each commodity, we then discard one-third of its words, specifically those in the bottom tercile by this measure

¹⁵ There were 5,459 subcommodities in 2002. The average number of subcommodities per commodity is 12; the minimum is 1; and the maximum is 154.

¹⁶ There are 250 such words including “commercial” and “component.” The Internet Appendix provides a full list.

(the least specific words).¹⁷ Overall we consider 7,735 commodity words. For instance, the last row of Table 1 illustrates that words such as film, projectors, photoengraving, and microfilm characterize the BEA commodity “photographic and photocopying equipment.”

The “Detailed Item Output” table also provides metrics of economic importance. We compute the *relative* economic contribution of a given subcommodity (ω) as the dollar value of its production relative to its commodity’s total production (see the last column of Table 1). Each word in a subcommodity’s textual description is assigned the same ω . Because a word can appear in several subcommodities, we sum its ω ’s within a commodity. A given commodity word is important if this fraction is high. We define the commodity-word correspondence matrix (CW) as a three-column matrix containing: a commodity, a commodity word, and its economic importance (e.g., {#333315, projector, 0.044} or {#333315, photolithography, 0.162}).

Next, we compute the intensity of vertical relatedness between pairs of commodities. We construct the sparse square matrix V based on the extent to which a given commodity is vertically linked (upstream or downstream) to another commodity. From the Make table, we create $SHARE$, an $I \times C$ matrix (Industry $\times$ Commodity) that contains the percentage of commodity c produced by a given industry i . The USE matrix is a $C \times I$ matrix that records the dollar value of industry i ’s purchase of commodity c as input. The $CFLOW$ matrix is then given by $USE \times SHARE$, and is the $C \times C$ matrix of dollar flows from an upstream commodity c to a downstream commodity d . Similar to Fan and Goyal (2006), we define the $SUPP$ matrix as $CFLOW$ divided by the total production of the downstream commodity d . $SUPP$ records the fraction of commodity c that is used as an input to produce commodity d . Similarly, the matrix $CUST$ is given by $CFLOW$ divided by the total production of the upstream commodities c , and it records the fraction of commodity c ’s total production that is used to produce its commodity d . The V matrix is then defined as the average of $SUPP$ and $CUST$. A larger element in V indicates a stronger vertical relationship between commodities c and d .¹⁸ Note that V is sparse (i.e., most commodities are not vertically related) and is nonsymmetric as it features downstream ($V_{c,d}$) and upstream ($V_{d,c}$) directions.

Figure 1 presents a snapshot of the direction and intensity of upstream and downstream vertical relatedness associated with the “photographic and photocopying equipment” commodity (#333315). As measured by V , this commodity is downstream to “semiconductor and related device manufacturing” (#334413) and “coated and laminated paper, packaging paper, and plastics film manufacturing” commodities (#32222A), which supply 2.2% and 1.4% of their respective production to the “photographic and photocopying

¹⁷ Hoberg and Phillips (2010) use a similar tercile approach to discard the most broad/vague words.

¹⁸ Alternatively, we consider in unreported tests the maximum between $SUPP$ and $CUST$, and also $SUPP$, or $CUST$ alone, to define vertical relatedness. Our results are robust.

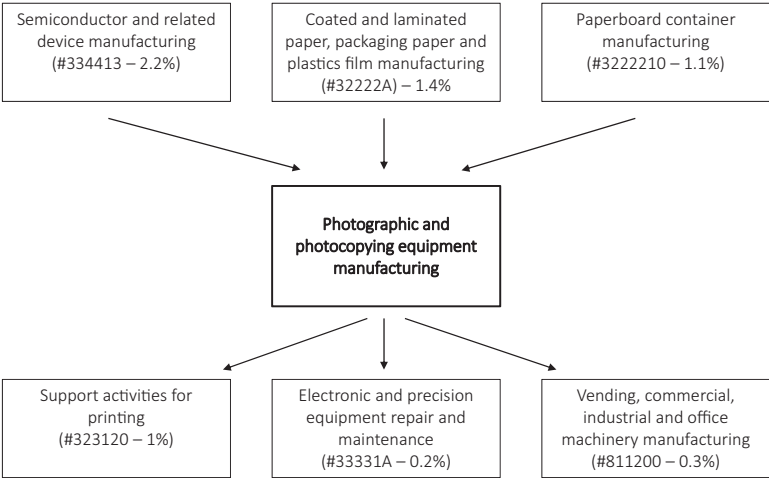


Figure 1
Example of BEA commodity-commodity vertical relatedness (the *V* matrix)

This figure presents a snapshot of the direction and intensity of upstream and downstream vertical relatedness between commodities based on the 2002 BEA input-output tables (the *V* matrix), focusing on the “Photographic and photocopying equipment manufacturing” commodity (#333315). We display some upstream and downstream commodities and the intensity of their vertical relatedness based on the elements of the matrix *V*.

equipment” commodity. This commodity is itself upstream to the “support activities for printing” (#323120) and “electronic and precision equipment repair and maintenance” (#33331A) commodities, supplying 1% and 0.2% of its production to these commodities.

2.3 Text-based vertical relatedness

We combine the vocabulary in firm 10-Ks and the vocabulary defining the BEA IO commodities to identify vertical relatedness between firms. The intuition is as follows. Because vertical relatedness is observed from the IO tables at the IO commodity level (see description of the matrix *V* above), we can score every pair of firms *i* and *j* based on the extent to which they are located upstream or downstream to each other by (1) mapping *i*’s and *j*’s text to the subset of IO commodities it provides, and (2) determining the intensity of *i* and *j*’s vertical relatedness using the vertical relatedness matrix *V*.

We proceed in three steps. First, we link each firm-year in the Compustat/Edgar universe to BEA IO commodities by computing the similarity between the given firm’s business description and the textual description of each BEA commodity. We limit attention to words that appear in the Hoberg and Phillips (2016) post-processed universe based on firms’ 10-Ks. Although uncommon, a firm will sometimes mention its customers or suppliers in its 10-K’s product description. For example, a coal manufacturer might mention that its products are “sold to” the steel industry. To ensure that vocabulary from firms’ 10-K product descriptions is not contaminated by such vertical

links, we remove any mentions of customers and suppliers using eighty-one typical phrases listed in the Internet Appendix.¹⁹ We then represent both firm vocabularies and BEA commodity vocabularies as vectors having a length equal to the number of nouns and proper nouns appearing in 10-K business descriptions in each year (63,367 in 1997, e.g.), with each element corresponding to a single word. By representing BEA commodities and firm vocabularies as vectors in the same space, we are able to assess firm and commodity similarity.

Our next step is to compute the “firm to IO commodity correspondence matrix” B . This matrix has dimension $M \times C$, where C is the number of IO commodities, and M is the number of firms. An entry $B_{m,c}$ (row m , column c) is the cosine similarity of the text in the given IO commodity c , and the text in firm m ’s business description. In this cosine similarity calculation, commodity word vector weights are assigned based on the words’ economic importance from the CW matrix (see above), and firm word vectors are equally-weighted following Hoberg and Phillips (2016). We use cosine similarity because it controls for document length, is bounded in $[0,1]$, and is well established in computational linguistics (see Sebastiani 2002). The matrix B thus indicates which IO commodities a given firm’s products are most similar to. Intuitively, the product description of a firm (say m) that manufactures and sells photocopying equipment will appear largely similar to the text for the “photographic and photocopying equipment” commodity (say c), so that the element $B_{m,c}$ will be large.

We then measure the extent to which firm i is located upstream relative to firm j using the following triple product:

$$UP_{i,j} = [B \cdot V \cdot B']_{i,j}, \quad (1)$$

which results in an $M \times M$ matrix of upstream-to-downstream relatedness between firms i to firms j . Note that direction is important, and the UP_{ij} matrix is not symmetric. Upstream relatedness of i to j is thus the i ’th row and j ’th column of this matrix. Firm-pairs receiving the highest scores for vertical relatedness are those having vocabulary that maps most strongly to IO commodities that are vertically related to matrix V (constructed only using BEA relatedness data). Thus, firm i is located upstream from firm j when i ’s business description is strongly associated with commodities used to produce other commodities whose description resembles firm j ’s product description. Downstream relatedness is simply the mirror image of upstream relatedness, $DOWN_{i,j} = UP_{j,i}$. We repeat this procedure for every year and create a time-varying network of firm-pairwise vertical relatedness. In section 6, we also use the diagonal entries of the triple product ($UP_{i,i}$) to measure the extent to which a given firm is “vertically integrated” based on whether its business description contains many word pairs that are vertically related.

¹⁹ We think this step is important, but our results are robust if we exclude this step.

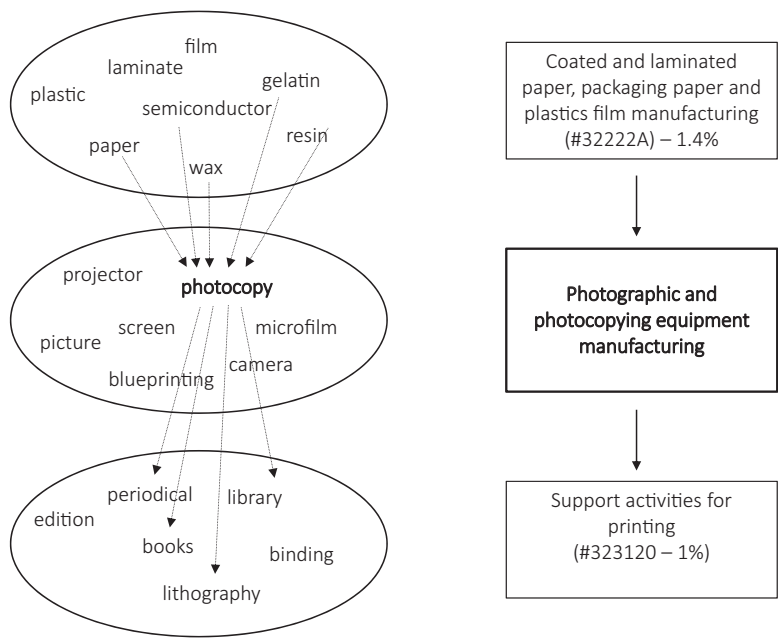


Figure 2
Vertical relatedness between words based on BEA

This figure displays how we use vertical relatedness between commodities (on the right) to compute vertical relatedness between words in the corresponding commodity vocabularies. For instance, the word “photocopy” (part of the vocabulary of our previous example) is downstream of “plastic,” “semiconductor,” and “resin,” but upstream of “periodical,” “book,” and “library.”

Figures 2 and 3 illustrate the intuition for the triple product $B \cdot V \cdot B'$. Figure 2 depicts how we use vertical relatedness between commodities (on the right) to compute vertical relatedness between words in the corresponding commodity vocabularies. For instance, the word “photocopy” (part of the vocabulary of our previous example) is downstream relative to “plastic,” “semiconductor,” and “resin,” but upstream relative to “periodical,” “book,” and “library.” Figure 3 depicts some words extracted from the 10-K business description of two sample firms, key to constructing the B matrices in Equation (1). Our measure of vertical relatedness thus intuitively uses the vertical word mappings from the BEA data (as in Figure 2), and the words in firm business descriptions to map specific firm-pairs to the BEA vertically related vocabularies. The arrows indicate vertical relatedness between words, highlighting that the firms A and B have nontrivial vertical relatedness ($UP_{A,B}$ is large), as firm A’s business description contains many words that are upstream relative to many words in firm B’s business description.

As an illustration, Table 2 presents a snapshot of the UP_{ij} pairwise vertical network in 2010, focusing on two upstream and downstream companies. The upper part of the table displays a representative set of companies that are

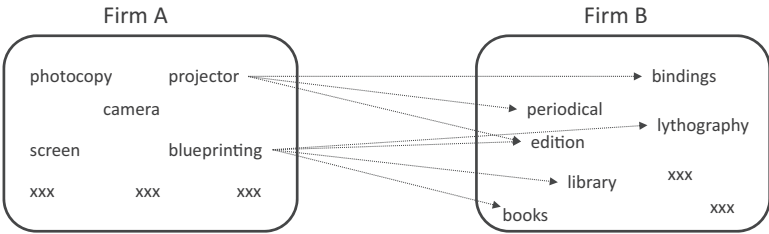


Figure 3
Illustration of firm-pair vertical relatedness ($UP_{A,B}$)

This figure illustrates some words extracted from the 10-K business description of two sample firms, key to constructing the B matrices in Equation (1). Our measure of vertical relatedness uses the vertical word mappings from the BEA data (as in Figure 2) and the words in firm business descriptions to map specific firm-pairs to the BEA vertically related vocabularies. The arrows indicate vertical relatedness between words, highlighting that the firms A and B have nontrivial vertical relatedness ($UP_{A,B}$ is large).

upstream to Tesla, an electric car manufacturer. These includes companies selling auto parts (e.g., Sorl Auto Parts), metals (e.g., Titanium Metals), electronic instruments (e.g., Cabot Microelectronics), or hydraulics products (e.g., Sun Hydraulics). In the bottom part of the table, we display a set of companies that are downstream from Encore Wire, a manufacturer of copper and aluminum wires. Downstream-located companies includes firms active in the car (e.g., General Motors), electronics (e.g., Pulse Electronics), communication (e.g., Atheros Communications), energy (e.g., Massey Energy), or pumps (e.g., Gorman-Rupp) business.

It is important to note our text-based measure of pairwise vertical relatedness is designed to capture the extent to which two firms operate in vertically related product markets (e.g., Encore Wire and Caterpillar from Table 2). It does not measure actual shipments (if any) between pairs of firms. This distinction is important in light of the recent analysis of Atalay, Hortacsu, and Syverson (2014) who document limited physical shipments within vertically integrated firms in the United States, and suggest that intangible assets are important drivers of vertical boundaries. Our new construct thus provides a broad measure of vertical relatedness, which captures the “potential” for vertical integration between firm pairs.

2.4 NAICS-based vertical relatedness

As a benchmark for evaluating our new text-based pairwise vertical network, we also consider the NAICS-based vertical relatedness network used in previous research (see, e.g., Fan and Goyal 2006). To compute firms’ pairwise vertical relatedness based on NAICS, we follow the literature and construct a matrix Z , which is analogous to the matrix V in the preceding section but is based on BEA industries instead of commodities. Following common practice in the literature, we map BEA industries to NAICS industries and use two numerical thresholds to identify meaningful levels of vertical relatedness between industries: 1% and 5%. A given industry i is deemed as upstream (downstream) relative to

Table 2
Vertical relatedness examples: Tesla and Encore Wire

Firm <i>i</i> (Upstream)	Firm <i>j</i> (Downstream)	Year	Vertical Score
MEASUREMENT SPECIALTIES INC	TESLA INC	2010	0.02294
EDAC TECHNOLOGIES CORP	TESLA INC	2010	0.01273
DIODES INC	TESLA INC	2010	0.01276
FUEL SYSTEMS SOLUTIONS INC	TESLA INC	2010	0.02227
SORL AUTO PARTS INC	TESLA INC	2010	0.02293
CLEANTECH SOLUTIONS INTL INC	TESLA INC	2010	0.01988
SUN HYDRAULICS CORP	TESLA INC	2010	0.02070
ENCORE WIRE CORP	TESLA INC	2010	0.02005
CABOT MICROELECTRONICS CORP	TESLA INC	2010	0.01567
LITTELFUSE INC	TESLA INC	2010	0.01466
TITANIUM METALS CORP	TESLA INC	2010	0.02239
COOPER TIRE & RUBBER CO	TESLA INC	2010	0.01696
HONEYWELL INTERNATIONAL INC	TESLA INC	2010	0.02560
GENERAL ELECTRIC CO	TESLA INC	2010	0.01275
...	...	...	...
ENCORE WIRE CORP	ALLIED MOTION TECHNOLOGIES	2010	0.01425
ENCORE WIRE CORP	MASSEY ENERGY CO	2010	0.01302
ENCORE WIRE CORP	BEACON ROOFING SUPPLY INC	2010	0.01452
ENCORE WIRE CORP	GENERAL MOTORS CO	2010	0.01866
ENCORE WIRE CORP	CATERPILLAR INC	2010	0.01692
ENCORE WIRE CORP	TESLA INC	2010	0.01445
ENCORE WIRE CORP	PULSE ELECTRONICS CORP	2010	0.01945
ENCORE WIRE CORP	UNIVERSAL DISPLAY CORP	2010	0.01260
ENCORE WIRE CORP	GORMAN-RUPP CO	2010	0.02346
ENCORE WIRE CORP	HARLEY-DAVIDSON INC	2010	0.01379
ENCORE WIRE CORP	ADVANCED BATTERY TECH INC	2010	0.01695
ENCORE WIRE CORP	ATHEROS COMMUNICATIONS INC	2010	0.01553
ENCORE WIRE CORP	STEEL DYNAMICS INC	2010	0.02823
...	...	...	...

This table provides examples of vertical relatedness for two illustrative companies in 2010 based on the network matrix UP_{ij} , defined in Section 2. The table displays the upstream (i) and downstream (j) firms, the year, and the vertical score for each firm-pairs.

industry j when the flow of goods Z_{ij} (Z_{ji}) is larger than this threshold. To create a network of pairwise vertical relatedness between firms (henceforth NAICS-based network), we simply assign firms into NAICS industry based on the NAICS code reported by each firm in Compustat. Therefore, the degree of vertical relatedness between firms corresponds to that of their respective industries. We find that the 1% and 5% flow thresholds generate NAICS-based vertical relatedness networks that have granularity of 1.32% and 9.17%, which means that 9.17% of randomly chosen firm pairs are vertically related in this network. For simplicity, we label these vertical networks as “NAICS-1%” and “NAICS-10%”, respectively.

To ensure our textual networks are comparable, we choose two analogous textual granularity levels: 10% and 1%. These two text-based vertical networks define firm pairs as vertically related when they are among the top 10% and top 1% most vertically related firm-pairs using the textual scores. We label these networks as “Vertical Text-10%” and “Vertical Text-1%.” Note that the NAICS-based networks rely not only on the quality of NAICS industries to delineate product markets, but also on the mapping between NAICS and BEA industries.

Table 3
Vertical network statistics and validity tests

Network:	Vert. text-10%	Vert. text-1%	NAICS-10%	NAICS-1%	TNIC-3
<i>A: Networks' statistics</i>					
Granularity	10%	1%	9.17%	1.32%	2.36%
% of pairs in TNIC-3	1.52%	2.50%	2.87%	3.79%	100%
% of pairs in the same SIC	0.77%	1.01%	0.36%	0.20%	36.61%
% of pairs in the same NAICS	0.60%	0.59%	0.12%	0.11%	36.27%
% of pairs in the same SIC or NAICS	0.85%	1.07%	0.36%	0.20%	39.55%
% of pairs in Vert. Text-10%	100%	100%	10.59%	13.66%	7.17%
% of pairs in Vert. Text-1%	10%	100%	1.17%	1.17%	1.18%
% of pairs that include a financial firm	9.63%	1.87%	51.20%	36.97%	55.81%
% of (no fin.) pairs in Vert. Text-10%	100%	100%	20.42%	21.03%	11.63%
% of (no fin.) pairs in Vert. Text-1%	10%	100%	2.39%	1.85%	2.68%
<i>B: Actual supplier-customer relationships</i>					
Supp-cust pairs (Compustat) (38,734 pairs)	21.49%	6.36%	14.47%	5.09%	14.09%
Supp-cust pairs (CIQ) (12,293 pairs)	4.05%	0.76%	3.01%	0.35%	12.293

This table presents the characteristics of our five distinct networks: Text-10% and Text-1% networks correspond to vertical networks based on textual analysis set at a 10% and, respectively, 1% granularity level, NAICS-1% and NAICS-5% correspond to vertical networks based in the 2002 BEA Input-Output Table with relatedness cutoffs of 1% and 5%, respectively, and TNIC corresponds to the horizontal text-based Network Industry Classification developed by Hoberg and Phillips (2010). Panel A presents the fraction of firm-pairs in each network that are also present in other networks. Panel B presents the fraction of firm-pairs in each network that are also present in the networks of actual customer-supplier linkages, based on Compustat segments and Capital IQ client announcements.

Both industry classifications suffer from the same time-fixed and all-or-nothing limitations of SIC codes noted in Hoberg and Phillips (2016). A key advantage of the textual vertical network is that it relies on neither, and instead relies on the more detailed BEA commodity space. Unlike existing measures, our text-based measure of vertical relatedness generates a set of vertically related peers that is customized to each firm's unique product offerings, and varies over time as firms dynamically update their product descriptions.

2.5 Vertical network statistics and validation

Panel A of Table 3 presents comparative statistics for four vertical and one horizontal relatedness networks: Vertical Text-10%, Vertical Text-1%, NAICS-10%, NAICS-1%, and the TNIC-3 network developed by Hoberg and Phillips (2016). The first four capture pairwise vertical relatedness, and the TNIC-3 network captures pairwise horizontal relatedness. The first row shows that the NAICS-10% and NAICS-1% networks have granularity levels of 9.17% and 1.32%, respectively. These levels, by design, are comparable to the 10% and 1% levels for the "Vertical Text-10%" and "Vertical Text-1%" networks.

Reassuringly, the second to fourth rows show that the four vertical networks overlap little with the horizontal TNIC-3 network, and horizontal networks based on firms' classified in the same SIC or NAICS industries. Thus, horizontal contamination is low. However, the fifth and sixth rows illustrate that the vertical

networks are also quite different. Only 10.59% of firm-pairs in the NAICS-10% network overlap with the Vertical Text-10% network, and only 1.17% of firm-pairs overlap for the Vertical Text-1% and NAICS-1% networks.

A key reason for this difference relates to financial firms. The eighth row shows that financial firm pairs are rarely classified as vertical by the text-based vertical networks (9.63% and 1.87% of linked pairs for the 10% and 1% networks). In contrast, financial firms account for a surprisingly large 51.20% and 39.97% of firm-pairs in the NAICS-based vertical networks. When we discard financials, the last two rows show that overlap between our text-based and NAICS-based networks roughly doubles. Because theories of vertical integration are based on nonfinancial issues, such as relationship-specific investment and ownership rights, these results further validate the use of our text-based network.

We perform several analyses to assess the external validity of our vertical relatedness network. First, we examine the extent to which our measure predicts actual vertical relationships between firm pairs. Similar to Barrot and Sauvagnat (2016) we rely on Compustat segment files to identify customer representing more than 10% of firms' sales. For our sample period, we obtain 38,734 supplier-customer pairs (excluding financials and utilities), corresponding to 4,689 unique suppliers and 2,306 customers.²⁰ Alternatively, we use new client announcements from Capital IQ's "Key Developments" database and obtain 12,293 announced supplier-customer pairs (4,474 suppliers and 4,459 customers) where both partners have available links to Compustat in our sample period.²¹ Panel B of Table 3 indicates that our Vertical Text-10% network captures more than 21% of the Compustat customer-supplier pairs, whereas only 14% are classified as vertically related using the NAICS-10% network. The text-based network is thus 50% more effective. A similar picture emerges when we focus on the Capital IQ new client announcements, where our text-based measure performs 34% better than the NAICS-based measure.

Second, we study how firms that are related vertically respond to trade credit shocks. Intuitively, firms operating in vertically related markets should experience *negatively* correlated shocks in accounts payables versus accounts receivables due to their supply chain adjacency. For example, a shock to an upstream market's *receivables* should be associated with a similar shock to the downstream market's *payables*. In contrast, firms operating in horizontally related markets should experience trade-credit shocks that are positively correlated. For brevity, we present the details of the test and results verifying

²⁰ Regulation SFAS No. 131 requires that firms disclose information about larger customers. We obtain customer-suppliers links using the replication kit of Barrot and Sauvagnat (2016) posted on the *Quarterly Journal of Economics* Web site.

²¹ Capital IQ records corporate announcements from over 20,000 sources including public news announcements, company press releases, regulatory filings, call transcripts, and company Web sites and filters these data to eliminate duplicates, identify the companies involved, and categorize each announcement into event categories (we focus on new clients). These data are available starting from 2002.

that these properties hold significantly for our text-based measures of vertical relatedness in the Internet Appendix. In contrast, these properties do not hold for the NAICS-based measure.

3. Innovation and Vertical Acquisitions

To test our main hypothesis, we study the determinants of vertical acquisitions. We focus on targets (i.e., the sellers of assets) as they are the party that loses control rights due to the transaction. Our baseline test focuses on the distinction between unrealized and realized innovation, and how both affect the likelihood of becoming a target in a vertical acquisition.

3.1 Sample and definition

We gather data on mergers and acquisitions from the Securities Data Corporation SDC Platinum database. We consider all announced and completed U.S. transactions with announcement dates between January 1, 1996 and December 31, 2013, that are coded as a merger, an acquisition of majority interest, or an acquisition of assets. As we are interested in situations where the ownership of assets changes hands, we only consider acquisitions that give acquirers majority stakes. Following the convention in the literature, we limit attention to publicly traded acquirers and targets, and we exclude transactions that involve financial firms and utilities (SIC codes between 6000 and 6999 and between 4000 and 4999). To distinguish between vertical and nonvertical transactions, we require that the acquirer and the target have available data in the merged Compustat/Edgar universe (see Section 2.2.1).

Panel A of Table 4 indicates that the sample consists of 4,974 transactions and tabulates how many of these transactions are classified as vertical using various networks. We find that 39% are vertically related using the Vertical Text-10% network, whereas just 13% are vertical according to the NAICS-10% network. Because both networks have similar granularity levels, it is perhaps surprising that the networks disagree sharply on the raw intensity of vertical acquisitions. Also, given the granularity of 10%, if transactions are random, we would expect to classify 10% of transactions as vertical. The fact that we find 39% is evidence that vertically acquisitions are relatively common. We also find that with both networks, vertical deals are almost evenly split between upstream and downstream transactions.

To further highlight differences between vertical networks, we examine whether transactions classified as vertical lead to actual increases in acquirers' vertical integration. We do so by searching for the terms "vertical integration" and "vertically integrated" in each firm's 10-K (excluding cases where the firm indicates that it is not integrated or lacks integration), and define a binary variable identifying when a firm indicates that it is vertically integrated (VI_{10k}). We then regress it on binary variables indicating whether a firm was a vertical or nonvertical acquirer in the last 3 years (as well as firm and year fixed effects).

Table 4
Mergers and acquisitions: Sample description

Measure:	All	Text-based		NAICS-based	
		Vertical	Nonvert.	Vertical	Nonvert.
Deal type:	(N)				
<i>A: Sample description</i>					
# Transactions	4,974	1,919	3,055	626	4,348
% Vertical (Nonvert.)		38.58%	61.42%	12.59%	87.41%
# Upstream		952		229	
# Downstream		967		320	
<i>B: Mentions of vertical integration after transactions</i>					
(<i>dep.var</i> : VI_{10k} (firm and year FE)		0.037** (0.018)	-0.014 (0.011)		
(<i>dep.var</i> : VI_{10k} (firm and year FE)				0.033 (0.026)	0.006 (0.011)
<i>C: Combined acquirers and targets returns</i>					
CAR(0)	0.54%	0.61%	0.49%	0.41%	0.55%
CAR(-1,1)	0.86%	0.94%	0.79%	0.40%	0.91%
# Transactions	4,629	1,807	2,822	580	4,049
<i>D: Examples of vertical deals</i>					
Acquirer	Target	Vert. score	Year	Value (\$ mio)	Type
Western Digital	Komag	0.0059	2007	985	Upstream
Conexant	GlobeSpanVirata	0.0042	2003	1,005	Downstream
General Electric	Edwards Systems	0.0241	2004	1,395	Upstream
3M Co	Cuno	0.0053	2005	1,612	Upstream
United Technologies	Sundstrand	0.0290	1999	4,408	Upstream
Thermo Electron	Fischer Scientific	0.0084	2006	10,291	Downstream

This table presents the characteristics of the transaction sample. Panel A displays statistics for vertical and nonvertical transactions (nonfinancial firms only). A transaction is classified as vertical if the acquirer and target are pairs in the Vertical Text-10% network or the NAICS-10% network. Panel B displays regression coefficients (and standard errors) from specifications regressing a binary variable indicating mentions of vertical integration in firms' 10-K (VI_{10k}) on binary variables indicating whether firms were vertical or nonvertical acquirers in the last 3 years, based on our the text-based or NAICS-based vertical networks. Specifications include firm and year fixed effects. Panel C displays the average cumulated abnormal announcement returns (CARs) of combined acquirers and targets, obtained from a market model estimated on 250 days prior to events. Panel D displays examples of large vertical deals (both upstream and downstream acquisitions), together with the vertical score between parties measures on the year of transaction. * $p > .1$; ** $p > .05$; *** $p > .01$.

Panel B of Table 4 indicates that vertical acquisitions identified using our text-based measure are followed by a significant increases in mentions of vertical integration (with a t -statistic of 2.04), while we detect no significant changes in the language associated with integration after vertical transactions identified using the NAICS-based measure (t -statistic of 1.27). These results suggest that the accumulated noise associated with NAICS-based vertical measure is substantial.

Panel C also displays the average abnormal announcement return (expressed as a percentage) of combined acquirers and targets in vertical and nonvertical transactions. We present these results mainly to compare with previous research (based on SIC or NAICS codes). Confirming existing evidence, combined returns across all transactions are positive and range from 0.54% to 0.86%. We also find that when vertical transactions are identified using our text-based measure, the combined returns are larger for vertical relative to nonvertical

Table 5
Summary statistics

Variable:	Mean	SD	Min	Max	#Obs.
R&D/sales	0.108	0.277	0.000	2.373	61,463
Patents/assets	0.009	0.028	0.000	0.186	61,463
log(assets)	5.680	1.913	1.780	10.338	61,463
log(age)	2.896	1.108	0.000	4.984	61,463
PPE/assets	0.259	0.225	0.006	0.898	61,463
Segments	1.477	0.936	1.000	12.000	61,463
MB	1.982	1.491	0.561	9.521	61,463
CF/assets	0.023	0.204	-1.048	0.366	61,463
OI/sales	0.047	0.359	-2.236	0.628	61,463
Sales growth	0.169	0.412	-0.545	2.324	61,463
HHI	0.237	0.198	0.016	1.000	61,463
End users	0.470	0.077	0.034	0.985	61,463
Target	0.065	0.246	0.000	1.000	61,463
Vertical target	0.024	0.153	0.000	1.000	61,463
Nonvertical target	0.041	0.198	0.000	1.000	61,463
corr(R&D/sales,Patents/assets)	0.321	—	—	—	61,436

This table displays summary statistics for the main variables used in the analysis. Appendix defines the variables.

transactions. This supports the idea that vertical deals are value creating as in Fan and Goyal (2006).

Finally, panel D presents six examples of large vertical acquisitions in our sample, representing both upstream and downstream purchases. Western Digital, a disk drive manufacturer, is an example of a downstream firm buying an upstream firm. They purchased Komag Inc. in 2007, a producer of thin-film discs used in hard drives. Another example of upstream integration is GE's acquisition of Edwards Systems Technology in 2004, which gave GE control of Edward's fire detection technology, used in GE's broad security offering.²² In contrast, Conexant, a semiconductor manufacturer, integrated downstream through the acquisition in 2003 of Globespan Virata Inc., a maker of broadband communications products that use semiconductor chips.

3.2 Measuring unrealized and realized innovation

To measure unrealized and realized innovation and map them to transactions, we merge the SDC transaction data with our merged Compustat/SEC EDGAR firm-year database. We discard financial firms and utilities, and firm-year observations with sales lower than \$5 million.²³ The resultant sample includes 61,463 firm-year observations over the period 1996-2013, corresponding to 8,283 distinct firms.

We empirically characterize firms' R&D intensity as unrealized innovation and patenting intensity as realized innovation. A firm's R&D intensity is the

²² Will Woodburn, CEO of GE Infrastructure, indicated that Edwards' technology will enable GE to offer an "integrated facility management solution to partners and end users." See Buildings (2004).

²³ The Internet Appendix shows that such a size threshold does not affect our conclusions.

dollar amount spent on R&D in a given year divided by sales.²⁴ A firm's patenting intensity is the number of patents granted in a given year divided by assets.²⁵ We obtain information on patent grants from the United States Patent and Trademark Office (USPTO) for each firm in our sample between 1996 and 2010 by combining information from the NBER patent data set and the recent data compiled by Kogan et al. (2017). We extend these data by manually matching all patent grants to firms in our sample between 2011 and 2013 using assignee names. We winsorize both variables (and all continuous variables used in the analysis) at the 1% and the 99% level, and the appendix defines the variables.

Table 5 presents summary statistics. The average R&D intensity is 0.108 with a standard deviation of 0.277. The average patenting intensity is 0.009 (i.e., one patent per year for each \$110 million of assets) with a standard deviation of 0.028. We also note that the correlation between firms' R&D and patenting intensity is 0.321. Finally, the odds of being purchased by a vertically related buyer in a given year is 2.4%, whereas the odds of a nonvertical acquisition is 4.1%. The other variables (e.g., size, age, or cash flow) are in line with existing research.

3.3 Main results

To explore the link between the stage of development of firms' innovation and vertical acquisitions, we begin by reporting summary statistics in Table 6 and find a large difference in the innovation characteristics of firms purchased in vertical and nonvertical deals. When compared to firms that participate in no acquisitions over our sample period (nonmerging firms), targets in vertical transactions spend significantly less on R&D and are granted more patents. In contrast, targets in nonvertical acquisitions spend more on R&D and are granted fewer patents. In panel B, each target (vertical and nonvertical) is directly compared to a matched firm with similar characteristics that did not transact in the past 3 years.²⁶ We find similar results.

We confirm these univariate results by estimating probit models in which the dependent variable is a binary variable indicating whether a given firm is a target in a vertical transaction in a given year. The main independent variables are the contemporaneous firms' R&D and patenting intensity. We include year fixed effects and cluster standard errors at the year and industry level based on

²⁴ Following Koh and Reeb (2015), we replace missing firm-year R&D intensity by the median value of the industry (based on TNIC-3 industries). We show in the Internet Appendix that our results are similar if we restrict our sample to firms that report nonmissing R&D or if we directly control for the intensity of missing firms' R&D as in Seru (2014).

²⁵ We focus on patents awarded by "grant year" as opposed to "application year" because our hypothesis concentrates on changes in investment incentives and holdup risk that materialize when realized innovation is legally protected. The use of grant dates also eliminates potential truncation bias (Lerner and Seru 2017).

²⁶ For every transaction, matched targets are the nearest neighbors from a propensity score estimation based on Fixed Industry Classification (FIC) industries (Hoberg and Phillips 2016) and firm size.

Table 6
Vertical acquisitions: Deal-level analysis

Variable:	R&D/sales	Patents/assets
A: Whole sample		
(i) Vert. targets	0.053	0.010
(ii) Nonvert. targets	0.136	0.009
(iii) Nonmerging firms	0.109	0.008
<i>t</i> -statistic [(i)-(ii)]	(-11.73)***	(0.69)
<i>t</i> -statistic [(i)-(iii)]	(-8.82)***	(3.25)***
<i>t</i> -statistic [(ii)-(iii)]	(4.21)***	(2.70)***
B: Matched targets		
(i) Vert. targets	0.053	0.010
(ii) Matched vert. targets	0.093	0.006
<i>t</i> -statistic [(i)-(ii)]	(-6.61)***	(4.66)***
(i) Nonvert. targets	0.136	0.009
(ii) Matched nonvert. targets	0.115	0.008
<i>t</i> -statistic [(i)-(ii)]	(2.76)***	(2.66)***

This table presents univariate comparisons of the R&D and patenting intensity of firms' acquired in vertical and nonvertical transactions, as well as firms that never participate in any transactions. Transactions are defined as vertical when the acquirer and target are in pairs in the Vertical Text-10% network. In panel A, we compare targets of vertical and nonvertical deals, and nonmerging firms. In panel B, each target is compared to a "matched" nonmerging target using a propensity score model based on industry, size, and year. The appendix contains the definitions of all the variables. We report *t*-statistics corresponding to tests of mean differences. * $p > .1$; ** $p > .05$; *** $p > .01$.

the FIC classification from Hoberg and Phillips (2016). We also standardize each independent variable by their sample standard deviation for convenience. Table 7 confirms that unrealized and realized innovation have opposite effects on firms' vertical boundaries. In the first column, the estimated coefficient on firms' R&D intensity is significantly negative, supporting our hypothesis that R&D-intensive firms are more likely to stay separate and retain residual rights of control. Separation is optimal when innovation is unrealized as it preserves ex ante incentives to maintain investments in innovation. In contrast, the coefficient on firms' patent intensity is positive and significant, supporting our prediction that realized innovation protected by patents fosters the benefits of integration, and increases the likelihood of vertical acquisitions as it mitigates risk of ex post holdup.²⁷

Following previous research on acquisition incidence (e.g., Hoberg and Phillips 2010), Column 2 adds several control variables to capture other potential determinants of vertical acquisitions. These include firm size, age, asset structure (the ratio of fixed assets and the number of business segments), profitability (cash flow over assets and operating income over sales) and growth prospects (market-to-book ratio and sales growth). We also include a control for product market concentration (HHI), and a text-based variable identifying the

²⁷ We report a similar estimation that identifies vertical acquisitions using the NAICS-based network (NAICS-10%) in the Internet Appendix. We continue to find our main results regarding R&D and patenting intensity on vertical acquisitions, albeit with materially weaker economic and statistical magnitudes.

Table 7
Determinants of vertical acquisitions

Dependent variable: Specification:	Prob(Vertical target)						
	Main (1)	Main (2)	Ind×Yr (3)	lags (4)	OLS (5)	Text (6)	Ind. (7)
R&D/sale	−0.247*** (0.05)	−0.092*** (0.03)	−0.038* (0.03)	−0.095*** (0.03)	−0.002*** (0.00)	−0.038* (0.02)	−0.253*** (0.03)
Patents/assets	0.063*** (0.01)	0.112*** (0.01)	0.074*** (0.01)	0.101*** (0.01)	0.005*** (0.00)	0.015* (0.01)	0.169* (0.01)
log(assets)		0.329*** (0.02)	0.246*** (0.02)	0.329*** (0.02)	0.018*** (0.00)	0.314*** (0.02)	0.331*** (0.02)
log(age)		0.079*** (0.01)	0.085*** (0.01)	0.065*** (0.01)	0.005*** (0.00)	0.087*** (0.01)	0.075*** (0.01)
PPE/assets		−0.015 (0.01)	−0.021 (0.02)	−0.003 (0.01)	−0.000 (0.00)	−0.020 (0.02)	−0.016 (0.01)
Segments		0.081*** (0.01)	0.086*** (0.01)	0.099*** (0.01)	0.011*** (0.00)	0.078*** (0.01)	0.080*** (0.01)
MB		−0.093*** (0.02)	−0.032* (0.02)	−0.075*** (0.02)	−0.001*** (0.00)	−0.072*** (0.02)	−0.060*** (0.02)
CF/assets		−0.022 (0.02)	−0.035* (0.02)	−0.015 (0.02)	−0.002*** (0.00)	−0.045** (0.02)	−0.047** (0.02)
OI/sales		−0.027 (0.02)	−0.012 (0.02)	−0.043* (0.03)	−0.002* (0.00)	0.005 (0.02)	−0.026 (0.02)
Sales growth		−0.099*** (0.02)	−0.063*** (0.02)	−0.046*** (0.02)	−0.003*** (0.00)	−0.102*** (0.02)	−0.087*** (0.02)
HHI		−0.026* (0.01)	−0.068*** (0.02)	−0.018 (0.02)	−0.001 (0.00)	−0.025* (0.02)	−0.040*** (0.01)
End users		−0.175*** (0.01)	−0.066*** (0.02)	−0.171*** (0.01)	−0.009*** (0.00)	−0.179*** (0.01)	−0.165*** (0.01)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ind×Year FE	No	No	Yes	No	No	No	No
#Obs.	61,463	61,463	58,257	52,257	61,463	55,833	61,463
Pseudo. R^2	.015	.129	.150	.127	.033	.123	.133

This table presents results from probit models in which the dependent variable is a dummy indicating whether the given firm is a target in a vertical transaction in a given year. Vertical transactions are identified using the Vertical Text-10% network. The first two columns are the baseline models without and with control variables. Column 3 includes industry × year fixed effect, where industries are defined using FIC-100 industries from Hoberg and Phillips (2016). Column 4 considers (1-year) lagged independent variables. Column 5 reports estimates from a linear probability model (OLS) instead of probit. Column 6 considers firms R&D and patenting intensities directly from 10-K mentions. In column 7, all independent variable are computed as industry (equally weighted) averages, based on TNIC-3 industries. Appendix defines the variables. The independent variables are standardized for convenience. All estimations include year fixed effects. Standard errors are clustered by FIC-300 industry and year and are reported in parentheses. * $p > .1$; ** $p > .05$; *** $p > .01$.

degree to which firms’ products exit the supply chain.²⁸ We continue to observe opposite effects of firms’ R&D and patenting intensity on vertical acquisitions.

The remaining columns in Table 7 indicate that our results for R&D and patenting intensity are robust. In Column 3, we add industry × year fixed effects (based on FIC-100 industries from Hoberg and Phillips 2016) to control for time-varying industry characteristics. In Column 4, we use lagged values for all independent variables. In Column 5, we estimate a linear probability

²⁸ We characterize whether a firm’s products or services exit the supply chain using the BEA exit correspondence matrix, which includes industries present in the Use table, but not in the Make table (retail customers, the government, and exports). A higher value for *End User* indicates that a firm has a higher fraction of its products and services sold to retail consumers, the government, or foreign entities.

model instead of the baseline probit. In Column 6 we measure firms' R&D and patenting intensity directly from firms' 10Ks, which avoids the well-known measurement problems of Compustat R&D reporting, and incomplete patent counts when assigning patents to firms (e.g., Lerner and Seru 2017). Specifically, we count the number of paragraphs mentioning R&D or patents in each 10K and scale by the total number of paragraphs. Finally, we follow Acemoglu et al. (2010) and consider industry averages (based on TNIC-3 industries) instead of firm-level independent variables. The main results hold in all cases.

Our main specification includes both firms' R&D and patenting intensities as independent variables, as this allows us to derive separate inferences for each, while holding the other constant. However, because these variables are positively correlated (0.321 in our full sample), we also examine if multicollinearity is a concern. We first examine the variance inflation factors based on our complete linear model in Column 5, and find that they are small (2.33 and 1.17) for both firms' R&D and patenting intensities. This suggests that multicollinearity is unlikely a problem in our setting. Second, we report in the Internet Appendix that our results continue to hold if we include firms' R&D and patenting intensity separately, if we consider specific subsamples in which the correlation between R&D and patenting is small, if we focus separately on upstream or downstream acquisitions, if we eliminate from the sample firms that are targeted in nonvertical acquisitions (thus comparing vertical acquirers to nonacquirers), if we focus on targeted firms (thus comparing vertical deals to nonvertical deals), and in a bootstrap analysis in which we reestimate our baseline specification 1,000 times on subsamples comprising 3,000 randomly selected firms.

3.4 Economic magnitudes

Because our main specification is a vertical acquisition probit model, we measure economic magnitudes using marginal effects ($dydx$), which capture the implied change in transaction incidence (dy) in response to a 1-standard-deviation change in each independent variable (dx). We also present the predicted transaction incidence for standard percentile breakpoints. Specifically, we first compute the predicted transaction incidence when all variables are set to their mean values. We then shift a given variable to its 25th percentile or 75th percentile (or to its 10th or 90th percentile) and recompute the predicted probability holding all other variables fixed. We thus report how predicted transaction probabilities shift as a variable moves across these breakpoints.

Table 8 confirms that our findings are economically meaningful. The estimated marginal effect of a 1-standard-deviation increase in firms' R&D intensity is associated with a 0.46% percentage point reduction in the probability of being acquired in a vertical transaction. This shift is economically large given that it represents one-third of the mean predicted probability of

Table 8
Implied economic magnitude

Measure:	<i>dydx</i> (1)	25 th (2)	Mean (3)	75 th (4)	10 th (5)	Mean (6)	90 th (7)
R&D/sales	−0.46%	1.49%	1.36%	1.33%	1.50%	1.36%	1.23%
Patents/assets	0.56%	1.24%	1.36%	1.51%	1.24%	1.36%	1.60%
log(assets)	1.65%	0.71%	1.36%	2.36%	0.43%	1.36%	3.93%
log(age)	0.39%	1.20%	1.36%	1.56%	1.03%	1.36%	1.78%
PPE/assets	−0.07%	1.41%	1.36%	1.34%	1.42%	1.36%	1.29%
Segments	0.41%	1.23%	1.36%	1.53%	1.23%	1.36%	1.90%
MB	−0.47%	1.57%	1.36%	1.31%	1.62%	1.36%	1.05%
CF/assets	−0.11%	1.37%	1.36%	1.33%	1.44%	1.36%	1.31%
OI/sales	−0.13%	1.37%	1.36%	1.33%	1.42%	1.36%	1.31%
Sales growth	−0.45%	1.52%	1.36%	1.31%	1.65%	1.36%	1.10%
HHI	−0.13%	1.43%	1.36%	1.33%	1.45%	1.36%	1.24%
End user	−0.88%	1.81%	1.36%	1.03%	2.32%	1.36%	0.78%

This table displays economic magnitudes associated with vertical acquisition incidence. All magnitudes are conditional and thus account for the effects of year and all control variables based on our baseline probit model (reported in Column 2 of Table 7). Column 1 reports marginal effects (*dydx*), which capture the implied change in transaction incidence (*dy*) in response to a 1-standard-deviation change in each independent variable (*dx*). The other columns present the implied variation around the unconditional transaction incidence. We compute the predicted value when all of the independent variables are set to their mean values. We then set each variable to its 25th percentile or 75th percentile in Columns 2 to 4 or to its 10th or 90th percentile in Columns 5 to 7 and recompute the predicted probability holding all other variables fixed. We then report how the predicted probability of being acquired vertically changes when a independent variable moves from its 25th (10th) percentile to its mean, and then to its 75th (90th) percentile. Appendix defines the variables.

1.36%. Because firms’ R&D intensity has a high degree of skewness, changing it from the 25th to the 75th percentile lowers acquisition incidence more modestly from 1.49% to 1.33%. The economic impact of firms’ patenting intensity has a similarly large 0.56% marginal effect, and acquisition incidence increases from 1.24% to 1.51% when patenting intensity moves from the 25th to the 75th percentile.

We also note that the economic magnitude of firms’ R&D and patenting intensities is large relative to other determinants of vertical acquisitions. For instance, the marginal effects of both are similar to those of the market-to-book ratio, sales growth, and age. They are also roughly one-third as large as the marginal effect of firm size, which is the most important driver of acquisitions. We conclude that the association between the stage of innovation and vertical acquisitions is large both nominally and relative to other variables.

4. Incentive Provision and Holdup Costs

To provide further support for our interpretation, we consider specific predictions regarding the economic mechanisms underlying our hypothesis. The theory of the firm emphasizes that, because of contract incompleteness, the integration of vertically related firms comes with both costs and benefits, so that vertical acquisitions occur as a trade-off between ex ante innovation incentives and ex post holdup risk. We thus test whether our results are indeed stronger in subsamples in which the importance of maintaining incentives and the risk of holdup are particularly elevated.

4.1 Internal provision of incentives

Grossman and Hart (1986) and Hart and Moore (1990) illustrate that the main cost of vertical integration operates through a reduction of incentives for the party that loses residual rights of control. We propose that integration through vertical acquisitions should be relatively less costly – and therefore more likely – when the reduction of innovation incentives is less important to the overall relationship. We propose that this is more likely when the acquirer, as the new owner of the residual control rights, has the ability to maintain investment incentives after the acquisition through compensation contracts and/or by keeping key employees.²⁹ We thus predict that firms' unrealized innovation should be a less important driver of their vertical acquisition when acquirers can better maintain innovation incentives post-acquisition.

To test this prediction we examine how the sensitivity of vertical acquisitions to firms R&D intensity varies with proxies for the difficulty to maintain innovation incentives within firms. Our first proxy is based on inventor mobility. Following Melero, Palomeras, and Wehrheim (2017), we track the careers of inventors listed on patents (from the PatentView initiative) using the disambiguation algorithm in Li et al. (2014). We focus on inventors who receive at least one patent grant with a public firm between 1990 and 2013. We then compute the fraction of inventors listed on patents in a given year who move to another firm in the next 5 years. We posit that higher inventor mobility reveals greater risk of key employee leaving the firm and hence more difficulty in maintaining innovation incentives ex post.³⁰

We develop two additional measures of the difficulty to maintain innovation incentives using the text in firm 10-Ks. First, we compute the fraction of a firm's 10-K paragraphs that contain at least one word from each of the following two lists: {employee*} and {contributions, retention, incentive, award*, equity, option, grant, deferred, vest*, stock, compens*, plan*, benefit*, shares}.³¹ Second, we identify key man insurance as the fraction of 10-K paragraphs containing the bigram “key man” (excluding “do not”). We propose that maintaining innovation incentives is relatively easier both for firms mentioning employee incentives and for firms less worried about losing key employees.

Table 9 reports how these proxies moderate the effect firms' R&D intensity on the probability that they are acquired vertically. We do so by adding interaction terms between each independent variable in our baseline (including year fixed effects) and an indicator variable identifying observations that have above median values of each moderating variable (which we label “*HIGH*”) defined

²⁹ We thank Oliver Hart for suggesting this interpretation of the theory of the firm.

³⁰ For example, Ouimet and Zarutskie (2016) show that acquirers can successfully retain some high value employees by offering them compensation increases.

³¹ ** indicates a wild card.

Table 9
Contract incompleteness and holdup costs

Dep. variable:	Prob(VERTICAL target)					
Proxies for:	Ex ante incentives			Ex post holdup		
HIGH based on:	Inventor mobility (1)	Employee incentives (2)	Keyman insurance (3)	Patent liquidity (4)	Patent infringement (5)	Number of peers (6)
R&D/sales	-0.065** (0.03)	-0.184*** (0.07)	-0.097*** (0.03)	-0.139** (0.08)	-0.124** (0.06)	-0.073*** (0.03)
R&D/sales × HIGH	-0.087** (0.05)	0.177*** (0.07)	0.097** (0.05)	0.041 (0.09)	0.031 (0.07)	-0.017 (0.06)
Patents/assets	0.104*** (0.02)	0.098*** (0.02)	0.109*** (0.01)	0.154*** (0.02)	0.160*** (0.02)	0.120*** (0.02)
Patents/assets × HIGH	-0.039 (0.03)	0.017 (0.03)	0.058 (0.06)	-0.070*** (0.03)	-0.062** (0.03)	-0.126** (0.06)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Controls × HIGH	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Year × HIGH FE	Yes	Yes	Yes	Yes	Yes	Yes
#Obs.	61,463	61,463	61,463	61,463	61,463	61,463
Pseudo. R ²	0.133	0.131	0.129	0.131	0.129	0.129
p-val. R&D/sales	0.024**	0.020**	0.021**	0.963	0.679	0.421
p-val. Patent/assets	0.799	0.169	0.858	0.134	0.042**	0.027**

This table presents results from probit models in which the dependent variable is a dummy indicating whether the given firm is a target in a vertical transaction in a given year. Vertical transactions are identified using the Vertical Text-10% network. The independent variables are similar to the baseline specification (Column 2 of Table 7), augmented with interaction terms between each independent variable (including the year fixed effects) and indicator variables (“HIGH”) identifying the upper half of the distribution of six different splitting variables. We define indicator variables and assign firm-year observation into groups every year. The splitting variables are inventor mobility (Column 1), employee incentives (Column 2), the presence of keyman insurance (Column 3), patent liquidity (Column 4), the intensity of patent infringement (Column 5), and the number of horizontal peers (Column 6). For brevity, we only report the coefficients on firms’ R&D and patenting intensity and their respective interactions. Appendix defines the variables. The independent variables and their interactions with HIGH are standardized for convenience. All estimations include year fixed effects and their interaction with HIGH. Standard errors are clustered by FIC-300 industry and year and are reported in parentheses. We also report (bottom of each column) the *p*-values corresponding to significance tests of the marginal effect of the interaction terms (see Ai and Norton (2003)). * *p* > .1; ** *p* > .05; *** *p* > .01.

separately in each year.³² For brevity, we only report our key R&D and patenting variables and the interaction terms. We also report (bottom of each column) the *p*-values corresponding to significance tests of the marginal effect of the interaction terms (see Ai and Norton 2003).

In the first column, we find a negative and significant interaction between *R&D/sales* and *HIGH*, which indicates that the negative association between firms’ R&D intensity and the odds of vertical acquisitions is significantly stronger for firms with higher inventor mobility. Separation is more likely for these firms when their innovation is still unrealized. Analogously, Columns 2 and 3 show that vertical acquisitions are less sensitive to firms R&D intensity when firms rank higher on their ability to provide incentives through contracts. Remarkably, and especially consistent with these tests pertaining to ex ante

³² To simplify interpretation, we scale each interaction term by its sample standard deviation.

incentives only, the interaction terms are only significant for firms' R&D intensity.

4.2 Holdup risk

As originally noted by Williamson (1979), the main benefit of vertical integration operates through a reduction in the risk of opportunistic behavior by the party holding residual rights of control. Vertical acquisitions and the resultant joint ownership of the realized innovation should thus be more likely when the risk of holdup is higher. On this ground, the likelihood that a firm is acquired vertically should be more sensitive to its realized innovation when the risk of holdup is higher for the other party.

We consider three variables indicating elevated holdup risk. First, we propose that patents with a more liquid secondary market (sales and licensing) are less prone to holdup risk, as the threat of selling the technology can limit opportunistic behaviors. We follow Serrano (2010) and Bowen (2016) and use the USPTO Patent Assignment Database (PAD) to compute the annual fraction of public firms' existing patents that are sold or licensed to other public firms between 1996 and 2013, and we average this rate over each firm's industry.³³

Second, we focus on the frequency with which patents are infringed in a given market (e.g., Tirole 2016), arguing that holdup risk is lower when intellectual property rights are less stringent and more often infringed. We measure the fraction of paragraphs in firms' 10-K that contain at least one word from each of the following two lists {patent*, patented} and {infringement, infringe}, and take the average in each industry and year. Third, we follow Acemoglu et al. (2010) and consider the number of firms active in each firm's TNIC-3 industry, and we propose that overall holdup risk is lower in industries populated by more firms.

Confirming the importance of holdup risk for our interpretation, Columns 4 to 6 of Table 9 indicate that the positive link between firms' patenting and vertical acquisitions is significantly smaller when the risk of holdup is lower. In particular, the negative coefficients on the interaction between *Patent/assets* and *HIGH* indicate that firms' patenting intensity is less important for vertical acquisitions in industries displaying an active secondary market for patent sales and licensing, industries where patent infringement is more common, and industries featuring many firms.

5. Alternative Explanations

Our results are consistent with the hypothesis that firms' vertical boundaries are sensitive to the stage of firms' innovation through the channels of innovation incentives and holdup risk. We recognize, however, that even our more refined

³³ We thank Don Bowen for sharing his Compustat-merged data on patent transactions from the PAD database.

results based on contracting, incentive and holdup measures do not fully rule out alternative explanations. To further explore robustness, we consider two specific alternatives.

First, the link between the stage of firms' innovation and vertical acquisitions could reflect a signaling hypothesis in which vertical buyers view patent grants as signals of innovative efficiency, thus triggering acquisitions. Second, our findings could be explained by "reverse-causality" as firms might respond to the likelihood of vertical acquisitions by simultaneously reducing their R&D intensity and increasing patents. We consider tests that limit the scope for these alternative explanations and reinforce our interpretation.

5.1 Innovation efficiency

First, we directly test whether our findings could reflect a link between vertical acquisitions and firms' innovative efficiency.³⁴ Under this hypothesis, vertical acquisitions are unrelated to the trade-off between ex ante incentives and ex post holdup risk, but reflect the fact that firms that produce more patents per unit of R&D expense are more likely to get acquired in vertical transactions, as potential buyers rely on patent grants as signals for efficiency in innovation activities. In that case, the contrasting effect of firms' R&D and patenting intensity on vertical acquisitions should be subsumed (or greatly weakened) when we control for firms' innovative efficiency.

To assess this possibility, we include proxies for innovation efficiency in our main specification. We follow Hirshleifer, Hsu, and Li (2013) and define innovative efficiency as the ratio of firms' patents per unit of R&D capital. Specifically, we scale the number of patents granted to firm i in year t by its 5-year (or 3-year) cumulated and depreciated R&D expenses (assuming a depreciation rate of 20%), computed at $t - 2$.³⁵ The first four columns of Table 10 present the results. For firm-year observations without any patent grants, we either replace missing values by zero in Columns 1 and 2, or exclude them from the sample in Columns 3 and 4. Across all specifications, we observe that the opposite effects of firms' R&D and patenting intensity on vertical acquisition odds are not affected by the inclusion of firms' innovative efficiency.

5.2 Nonvertical acquisitions

Second, we examine the link between the stage of development of innovation and *nonvertical* acquisitions. We focus on nonvertical transactions as control transactions since the issues of ex ante incentives, contracting difficulties, and potential holdup risk should be more specific to deals occurring among vertically related firms. For instance, unlike our prediction, theories of

³⁴ We note that waiting for the exact degree of efficiency can be consistent with our results as it may be too costly ex ante to fully contractually specify the degree of innovation efficiency and the effort needed to achieve that efficiency.

³⁵ That is, $Patents_{it} / (R\&D_{t-2} + 0.8R\&D_{t-3} + 0.6R\&D_{t-4} + 0.4R\&D_{t-5} + 0.2R\&D_{t-6})$.

Table 10
Innovation efficiency and nonvertical acquisitions

Dependent variable: Specification:	Prob(Vertical target)				Prob(target)	
	Main (1)	Main (2)	Main (3)	Main (4)	Nonvert. (5)	Horiz. (6)
R&D/sales	−0.092*** (0.03)	−0.088*** (0.03)	−0.162*** (0.06)	−0.095*** (0.03)	0.069*** (0.02)	0.090*** (0.02)
Patents/assets	0.111*** (0.01)	0.102*** (0.01)	0.079*** (0.02)	0.101*** (0.01)	0.002 (0.01)	0.002 (0.01)
Innov. efficiency (5yrs)	0.002 (0.01)		−0.036 (0.02)			
Innov. efficiency (3yrs)		0.002 (0.01)		−0.018 (0.02)		
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
#Obs.	61,463	61,463	21,768	25,187	61,463	61,463
Pseudo. R^2	0.129	0.129	0.139	0.146	0.050	0.061

This table presents results from probit models in which the dependent variable is a dummy indicating whether the given firm is a target in a vertical (or nonvertical) transaction in a given year. Vertical transactions are identified using the Vertical Text-10% network. The first four columns focus on vertical transactions and augment the baseline specification (Column 2 of Table 7) with controls for the efficiency of firms' innovation, measured over the last 5 or 3 years. In Column 5, the dependent variable is a dummy indicating whether the given firm is a target in a nonvertical transaction in a given year (identified using the Vertical Text-10% network.) In Column 6, dependent variable is a dummy indicating whether the given firm is a target in a horizontal transaction in a given year (identified using the TNIC-3 network). Both specifications include the same independent variable as the baseline specification (Column 2 of Table 7). For brevity, we only report the coefficients on firms' R&D and patenting intensity and their innovation efficiency. Appendix defines the variables. The independent variables are standardized for convenience. All estimations include year fixed effects. Standard errors are clustered by FIC-300 industry and year and are reported in parentheses. * $p > .1$; ** $p > .05$; *** $p > .01$.

acquisitions based on horizontal patent races predict that R&D-intensive firms have higher incentives to merge in order to internalize the effect of competition.³⁶

Column 5 of Table 10 present results of regressions in which the dependent variable is a dummy indicating whether a given firm is a target in a nonvertical transaction (acquirer-target pair is *not* in our vertical relatedness network) in the given year. Similar to Bena and Li (2014), and in sharp contrast to our results for vertical acquisitions, R&D-intensive firms are significantly more likely to become targets in nonvertical acquisitions. Mirroring the results in Bena and Li (2014), patenting intensity is not significantly related to nonvertical acquisitions. We obtain similar results in Column 6 for horizontal acquisitions (the acquirer and target are TNIC-3 industry rivals).

5.3 R&D and patenting intensity around acquisitions

The reverse causality hypothesis posits that firms decrease R&D and increase patents prior to an acquisition, perhaps to attract more favorable terms. Under this hypothesis, it is likely that R&D should bounce back after vertical acquisitions because the motive for its reduction was rent extraction, and

³⁶ For instance Phillips and Zhdanov (2013) predict and show empirically that R&D is positively related to transaction likelihood for horizontal transactions, as firms wish to internalize their R&D on competing products.

not maximization of operating value (which would be the objective post-acquisition). Our hypothesis based on the theory of the firm, in contrast, predicts that declines in R&D should be permanent post-acquisition. This permanently lower R&D further predicts that the increase in patents will be strong only around the acquisition date, as the reduced R&D eventually reduces patenting.

Contrasting these predictions empirically is challenging, because the majority of targets cease to exist as separate entities after their acquisition. Nevertheless, we can precisely track the evolution of R&D intensity around vertical transactions for a subsample of 731 targets that continue to operate independently for at least 1 year post-acquisition. We also construct a synthetic measure of R&D intensity by summing the targets' and acquirers' R&D prior to acquisition, and using the acquirer post-acquisition (e.g., Maksimovic, Phillips, and Prabhala 2011). With both approaches, we follow targets (and synthetic entities) from 3 years prior to their acquisitions to 3 years after. To track the evolution of firms' R&D and patenting intensity around transactions, we regress their R&D and patenting intensity on (seven) event-time binary variables identifying time to acquisitions (from $t - 3$ to $t + 3$).³⁷

Figure 4 plots the estimated event-time coefficients. In panel A, we find that targets' patenting activity peaks in the year of the acquisition and decreases afterward. The results are similar but less pronounced in panel B for synthetic entities. This evolution supports our hypothesis that vertical deals materializes when targets' innovation becomes realized and legally protected, which reduces holdup risk for acquirers. More importantly, target R&D intensity significantly declines post-acquisition in both panels, similar to Seru (2014). Although this evidence is indirect, the permanently lower R&D intensity post-acquisition is hard to reconcile with the reverse causality alternative. This is also inconsistent with an alternative story in which buyers purchase the most innovative firms and maintain their innovation incentives through internal contracting. Yet the observed decline in R&D intensity supports the idea that vertical deals take place between firms when preserving incentives for further development becomes relatively less important compared to owning property rights.

Overall, we present a large body of evidence and tests consistent with the idea that vertical transactions arise as an optimal trade-off between the need to mitigate holdup risk and preserve investment incentives. We acknowledge however that our evidence remains largely indirect due to the difficulty to observe and measure firms' incentives. In particular, we recognize that part of our results might arise because some acquirers use patent grants as indications for targets' genuine success (and not their innovation efficiency), triggering their acquisitions. This explanation can be distinct from our explanation if no *ex ante* incentives are needed for the effort needed to achieve success. Although various tests (especially our subsample tests based on proxies for

³⁷ Specifications include year and firm fixed effects. We require that each firm has at least one available observation before and after the deal.

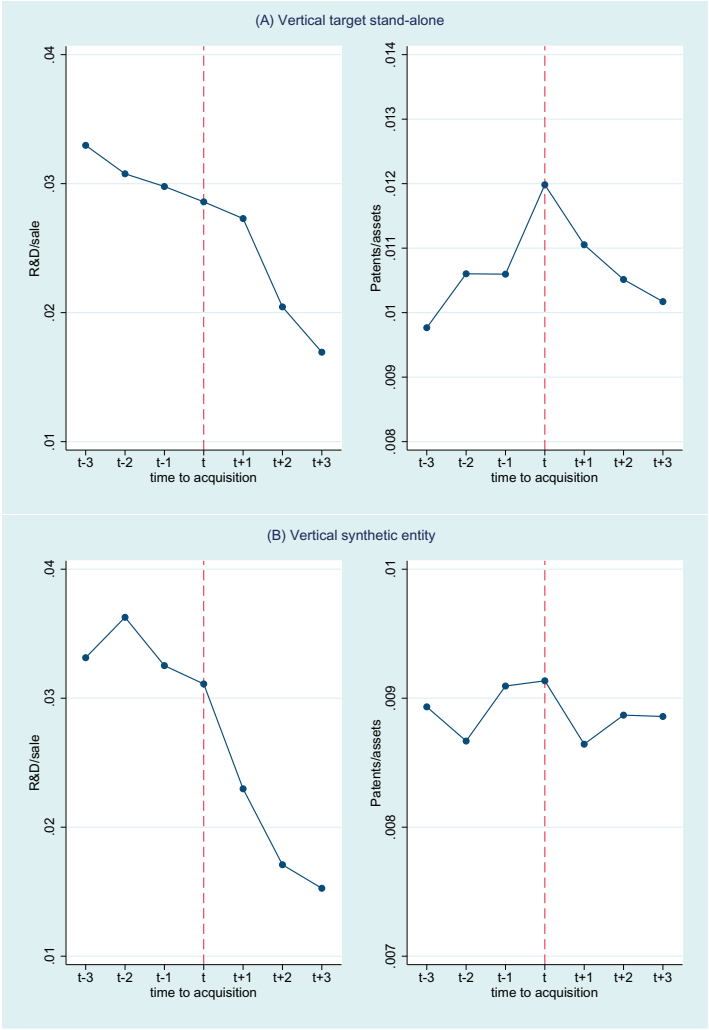


Figure 4
R&D and patenting intensities around vertical acquisitions.

This figure plots the evolution of targets' R&D and patenting intensity from 3 years prior to their vertical acquisitions to 3 years after. Panel A includes 731 targets that continue to operate independently for at least 1 year post-acquisition. In panel B, we construct a synthetic measure of R&D and patenting intensity by summing the targets' and acquirers' R&D and patents prior to acquisition, and using the acquirer post-acquisition. We track the evolution of firms' R&D and patenting intensity around transactions by regressing their R&D and patenting intensity on seven event-time binary variables identifying time to acquisitions (from $t-3$ to $t+3$) and report the estimated coefficients. We require that each firm has at least one available observation before and after the deal. Specifications include year and firm fixed effects.

incentive provision and holdup risk) limit support for explanations unrelated to incentives and holdup (such as the signaling of quality), we cannot fully rule out such alternatives.

6. Vertical Integration within Firms

To further examine the link between the stage of development of innovation and firms' vertical boundaries we also consider the degree of firms' vertical integration. While our central prediction is about the role of unrealized and realized innovation in vertical acquisitions (i.e., a reallocation of control rights between firms), the importance of ex ante innovation incentives and ex post holdup risk also should be prevalent in explaining variation in firms' vertical integration.

We define the extent to which a given firm i is "vertically integrated" (VI_i) using the diagonal entries of the matrix UP that we use to measure vertical relatedness between firms (see Equation (1) in Section 2.2.3). With that measure, a firm displays a higher degree of vertical integration if its business description contains many word pairs that are vertically related, that is, when its product vocabulary spans vertically related markets.³⁸ In the Internet Appendix, we present various evidence indicating that VI truly captures the extent to which firms are vertically integrated (e.g., correlation with mentions of vertical integration in firms' 10-Ks).

We first examine whether the degree of firms' vertical integration varies systematically across industries based on their respective innovative maturity. Our hypothesis predicts that vertical separation is preferred to integration when incentives for further innovation are important. Thus, industries in which innovation is mostly unrealized should feature independent (i.e., separate) firms displaying low levels of vertical integration, as these specialized firms will be integrated into other firms (via acquisitions) when their innovation is realized and patented. In contrast, industries where innovation is mostly realized should feature firms that display higher levels of vertical integration, resulting from their vertical acquisitions.³⁹

We assess these predictions by regressing firms' vertical integration on the average R&D and patenting intensity of their industry as well as (industry-level) controls variables that are similar to those used in our main acquisition specification (see Table 7). The first column of Table 11 reveals that firms in R&D-intensive industries are less vertically integrated, and those in patenting

³⁸ Unfortunately, data limitations prevent us from determining the economic weight of each product from firms' 10-K product descriptions. Thus, while VI is a novel measure that uniquely captures firm-level vertical integration, it cannot account for product-by-product importance.

³⁹ We do find that transactions classified as vertical are followed by an increase in our firm-level measure of vertical integration (VI). Using the Vertical Text-10% network, acquirers in vertical transactions experience a 12% increase in VI 1 year after the acquisition. In contrast, acquirers in nonvertical transactions experience an 8% decrease in VI .

Table 11
Determinants of vertical integration

	(Text-based) VI			
	Full sample	Full sample	At least one vertical acquisition	Never did a vertical acquisition
Dep. variable:	(1)	(2)	(3)	(4)
Ind.(R&D/sale)	−0.076*** (0.006)			
Ind.(Patents/assets)	0.031*** (0.006)			
Firm R&D/sale		−0.003 (0.003)	−0.006 (0.009)	−0.001 (0.004)
Firm patents/assets		0.007** (0.003)	0.035*** (0.012)	0.001 (0.002)
Controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	No	No	No
Firm FE	No	Yes	Yes	Yes
#obs.	61,463	60,463	11,328	49,135
Adj. R^2	.586	.849	.751	.859

This table presents results from OLS models in which the dependent variable is our firm-level measure of vertical integration *VI*. In all models, the control variables are similar to the baseline specification (Column 2 of Table 7). In Column 1 R&D and patenting intensities are computed at the industry-level, based on TNIC-3 industries, and the specification include year and industry fixed effects (defined using FIC-100 industries from Hoberg and Phillips (2016)). In Columns 2 to 4, we consider firms' R&D and patenting intensity, and specifications include year and firm fixed effects. Column 3 only includes firms that did at least one vertical acquisition during our sample period. Column 4 includes the complementary subsample: firms that did not do any vertical acquisitions during our sample. For brevity, we only report the coefficients on industry and firm R&D and patenting intensity. Appendix defines the variables. The independent variables are standardized for convenience. Standard errors are clustered by FIC industry and year and are reported in parentheses. * $p > .1$; ** $p > .05$; *** $p > .01$.

intensive industries are more integrated. These results confirms the importance of innovative maturity in explaining variation in integration across firms.

In Column 2 of Table 11 we examine how the variation of vertical integration *within* firms (over time) relates to their R&D and patenting intensity (through the inclusion of firm fixed effects in the specification). We report that changes in firms' degree of vertical integration is largely insensitive to their R&D intensity, consistent with our hypothesis which predicts that firms should remain independent and nonintegrated as long their innovation remains unrealized. In contrast, we observe a positive and significant link between firms' extent of vertical integration and their patenting intensity. Columns 4 and 5 reveal, however, that the positive relation between patenting intensity and integration is only present for firms that did acquire vertically related firms, suggesting that the positive integration-patent relationship stems from the patents obtained through vertical deals. Column 5 shows that the degree of vertical integration of firms not participating in vertical acquisitions is largely insensitive to the stage of development of innovation. Taken together, the observed variation in vertical integration within firms is consistent with the trade-off between ex ante innovation incentives and ex post holdup risk being at play to explain changes in vertical boundaries between firms, but not within firms.

7. Conclusions

Our paper examines the link between the stage of development of firms' innovation and vertical acquisitions. We consider theoretical predictions regarding how incentives to invest in R&D, and the potential for ex post holdup influence the incidence of vertical transactions. We develop a new measure vertical relatedness between firm pairs using computational linguistics analysis of firm product descriptions and how they relate to product vocabularies from the BEA Input-Output tables. The result is a dynamic network of vertical relatedness between publicly-listed firms, enabling us to precisely identify takeover transactions occurring between vertically related firms.

We find that unrealized innovation (R&D) and realized innovation (patents) are related to vertical acquisitions in opposite ways. R&D-intensive firms are less likely to be purchased by vertically related buyers, consistent with separation maintaining ex ante incentives to invest in intangible capital and to maintain residual rights of control, as in the property rights theory of Grossman, Hart, and Moore. In contrast, patenting intensive firms with high realized innovation are significantly more likely to be acquired by a vertical buyer, in line with legally enforceable residual rights of control reducing holdup risk that may can occur with commercialization and bringing new products to market, as in contracting costs literature of Williamson.

Appendix: Variable Descriptions

In this appendix, we describe the variables used in the analysis. We report Compustat items in parentheses when applicable. All ratios are winsorized at the 1% level in each tail.

New Data from Text Analysis

- VI_{10k} is a binary variable indicating whether a firm mentions the terms “vertical integration” and “vertically integrated” in its 10-K (excluding cases in which the firm indicates that lacks integration).
- VI measures the degree to which a firm offers products and services that are vertically related based on our new text-based approach to measure vertical relatedness (as defined in Section 6).
- $End\ user$ measures the degree to which a firm-year's products exit the U.S. supply chain. We characterize whether a firm supplies products or services that exit the supply chain using the exit correspondence matrix E , which includes industries present in the Use table, but not in the Make table (retail customers, the government, and exports). E is a one-column vector containing the fraction of each IO commodity that flows to these final users. We compute $End\ User_i$ as $[B \cdot E]_i$ (in the $[0, 1]$ interval) to compute the cosine similarity between the text in a firm's business description and the text in IO commodities that exit the supply chain. A higher value of $End\ User_i$ indicates that firm i has a higher fraction of

its products and services that are sold to retail, the government, or foreign entities.

- *Employee incentives* measures the fraction of paragraphs in a firm's 10-K that contain at least one word from each of the following two lists: {employee*} and {contributions, retention, incentive, award*, equity, option, grant, deferred, vest*, stock, compens*, plan*, benefit*, shares}.⁴⁰
- *Keyman insurance* measures the fraction of paragraphs in a firm's 10-Ks that contain the bigram "key man" (excluding "do not").
- *Patent infringement* measures the fraction of paragraphs that contain at least one word from each of the following two lists. List 1 is {patent, patents, patented}. List 2 is {infringement, infringe}.

Data from Existing Literature

- *R&D/sales* is equal to research & development expenses (XRD) scaled by the level of sales (SALE). We replace missing values by the median value of firms' industry (based in TNIC-3 industries).
- *Patents/assets* is the number of patents granted in a given year scaled by the level of assets (AT). Patents granted by the U.S. Patent and Trademark Office (USPTO) between 1996 and 2010 are obtained by combining information from the NBER patent data set and the recent data compiled by Kogan et al. (2017). We extend these data by manually matching all patent grants to firms in our sample between 2011 and 2013 using assignee names.
- *PPE/assets* is equal to the level of property, plant, and equipment (PPENT) divided by total assets (AT).
- *HHI* measures the degree of concentration (of sales) within TNIC-3 industries. We compute HHI as the TNIC-3 HHI in Hoberg and Phillips (2016), which is based on the text-based TNIC-3 horizontal industry network.
- *log/assets* is the natural logarithm of the firm assets (AT).
- *log(age)* is the natural logarithm of one plus the firm age. Age is computed as the current year minus the firm's founding date. When we cannot identify a firm's founding date, we use its listing vintage (based on the first year the firm appears in the Compustat database).
- *Segments* is the number of operating segments observed for the given firm in the Compustat segment database. We measure operating segments based on the NAICS classification.
- *MB* is the firm's market-to-book ratio. It is computed as total assets (TA) minus common equity (CEQ) plus the market value of equity ((CSHO×PRCC_F)) divided by total assets.

40 ** indicates a wild card.

- *CF/assets* is equal to income before extraordinary items (IB) plus depreciation and amortization (DP), scaled by total assets (AT)
- *OI/sales* is equal to operating income before depreciation (OIBDP) divided by sales (SALE).
- *Sales growth* is equal to annual growth rate of sales (SALE).
- *Inventor mobility* is the fraction of inventors listed on patents granted to a firm in a given year who move (i.e., listed on patents granted) to another firm in the next 5 years (from the PatentView initiative). We restrict attention to inventors who receive at least one patent grant with a public firm between 1990 and 2013, and exclude moves between firms due to mergers and acquisitions, based on transactions data from SDC.
- *Patent liquidity* is the annual fraction of public firms' existing patents that are sold or licensed (based on the USPTO Patent Assignment Database (PAD)) to other public firms between 1996 and 2013, averaged over each firm's industry (based on TNIC-3 industries).
- *Peers* measures the number of firms in each firm's TNIC-3 horizontal network.
- *Innovation efficiency* is equal the number of patents granted to given firm in year t divided by its 5-year (or 3-year) cumulated and depreciated R&D expenses (assuming a depreciation rate of 20%), computed at $t - 2$ as $R \& D_{t-2} + 0.8R \& D_{t-3} + 0.6R \& D_{t-4} + 0.4R \& D_{t-5} + 0.2R \& D_{t-6}$.

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